42. DEVELOPMENT OF THE VERTEBRAE

DURATION : 06:41

STUDY SHEET

COURSE SUMMARY

This video delves deeply into the development of the vertebrae and the musculoskeletal system, detailing the segmentation that influences the formation of the axial skeleton, ribs, and limbs. It explores the establishment of the notochord and neural tube, crucial elements for embryonic structuring, as well as the interactions between the ectodermal ring and these structures. Key concepts include differential growth, the formation of intervertebral discs, vertebral bodies, and the importance of information from the neural tube and notochord in this process. Practically, therapists will learn how to palpate and work on these different structures to influence vertebral development and identify potential clinical problems associated with dermalgia.

VERTEBRAL DEVELOPMENT

The development of the **locomotor system**, and more specifically of the **vertebrae**, relies on precise segmentation. This segmentation influences the trajectories of the **axial skeleton**, **ribs**, and **extremities**.

The establishment of the **notochord** induces the formation of the **neural tube**, which progressively organises along the longitudinal axis and structures the vascular system. This process is accompanied by **differential growth** that becomes anterior, leading to a flexion movement. The anterior part creates what is called the **ectodermal ring**, an epiblast tissue that flexes, thus forming this ring.

This flexion is essential because it leads to significant changes for the organism, particularly the phase of **embryo delimitation**. The presence of the neural tube and **neural crests** is crucial, as these elements interact with the ectodermal ring, providing important information to the notochord.

The movement of the ectodermal ring generates compression that influences development around the notochord. This process is linked to the formation of the **intervertebral discs** and **vertebral bodies**. The upper part transforms into a disc, while the lower and upper parts begin to form the vertebral body, under the influence of the notochord and neural crest.

The notochord, a remnant of the **nucleus pulposus**, plays a key role in the formation of vertebral bodies from the **sclerotome**, derived from the somites. Thus, the notochord induces the formation of the vertebral body and the vertebral disc, while surrounding structures, such as the arch and spinous process, are also influenced by the notochord and neural tube.

There are two types of information in the formation of a vertebra. For the spinous process, the information comes directly from the neural tube, while for the disc, it is linked to the notochord. The notochord also induces the formation of the **vertebral chlorines** originating from the sclerotome, while the neural tube contributes to the formation of the surrounding mesoblast.

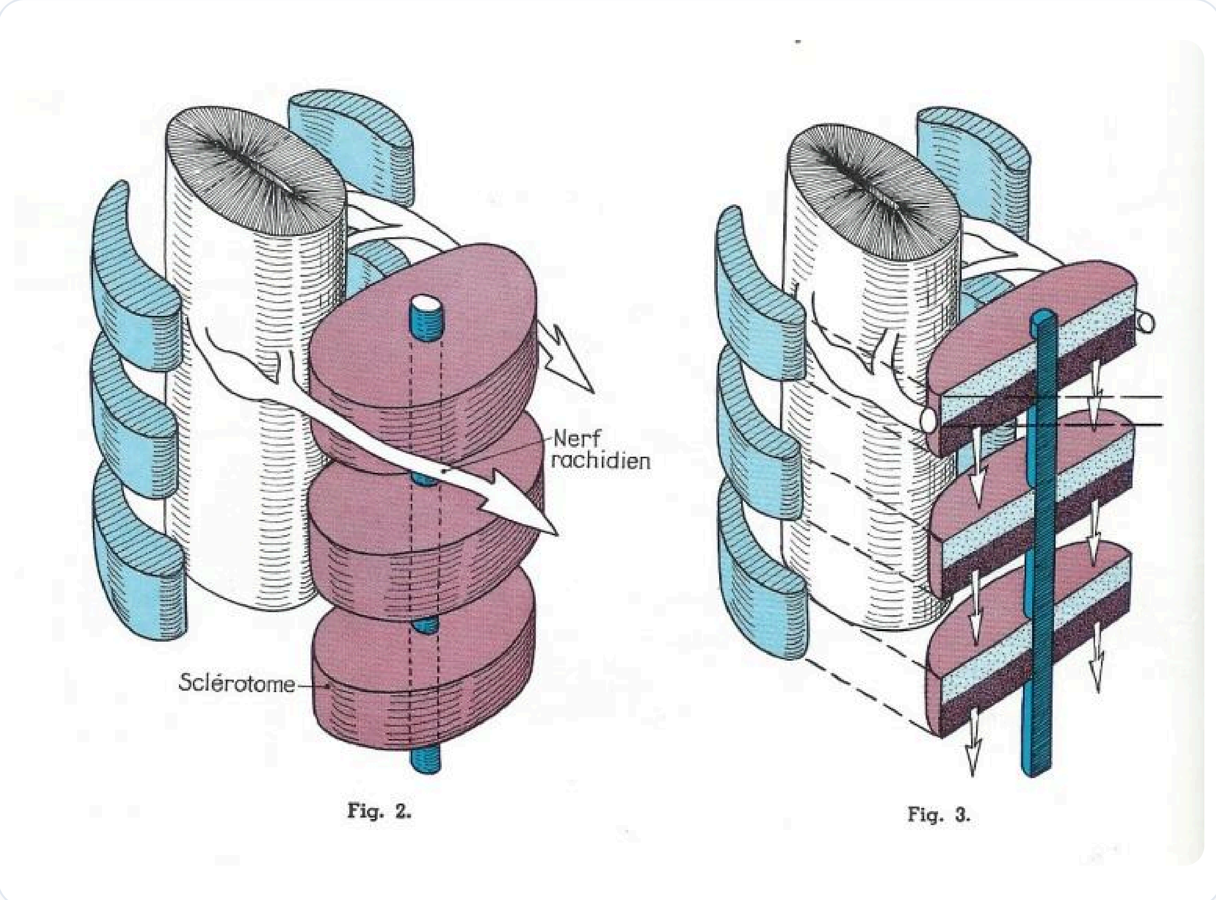

FIG. — Sclerotome, vertebrae, nerves

In complete formation, we observe the establishment of somites to form the sclerotome around the neural tube and notochord. The sclerotome is primarily responsible for the formation of the vertebral body, while a form of mesoblastic sclerotome contributes to the vertebral arch and spinous process.

The formation of the vertebral body and the vertebral arch occurs according to two planes: longitudinal for the vertebral body and transverse for the arch. This movement is essential for vertebral development.

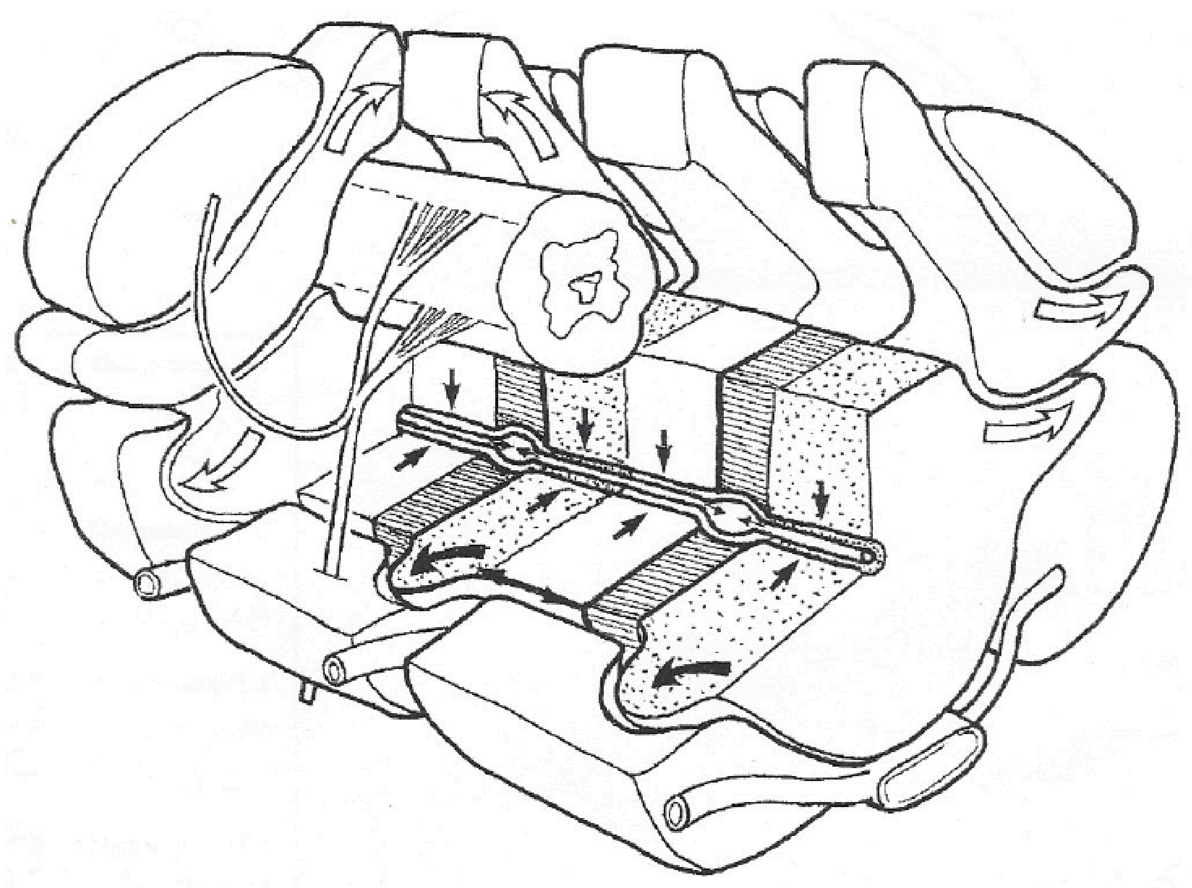

FIG. — Spinal CSF circulation

In practice, touching the body can be done at different levels: either at the level of the neural tube to work on the posterior arch, or on the longitudinal axis for the vertebral body, or in another plane for neural crest migration.

Each element, such as the skin, neural tube, and notochord, sends specific molecules to guide vertebral development. The notochord focuses on vertebra formation, while the neural tube deals with the arch and spinous process.

When development intensifies, it triggers specific inductions for the neural tube, leading to the formation of **dermatomes**, which are segmented throughout the body. **Dermalgia** on a dermatome can indicate a link with a viscus, while multiple dermalgias on the same dermatome may suggest a problem at the corresponding vertebra.

For example, dermalgia at level 9 can be associated with structures such as the **gallbladder**, **pancreas**, or **pyloric system**. In these cases, it is relevant to explore the associated vertebra to better understand the clinical implications.

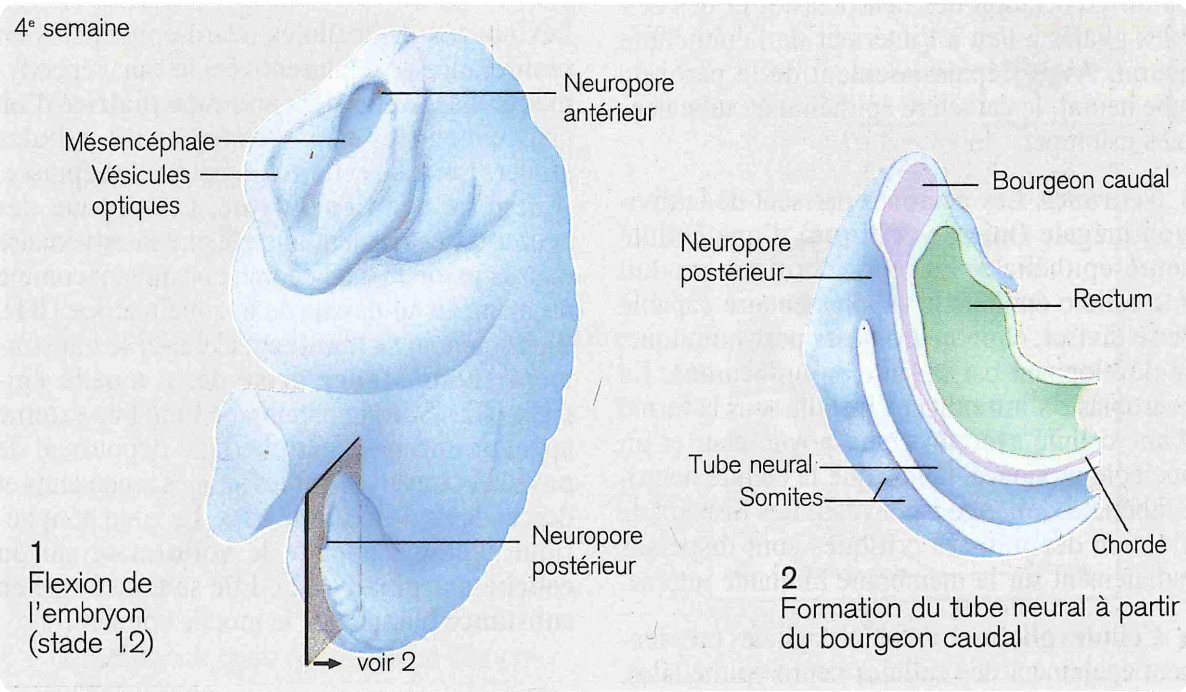

FIG. — Embryo 4th week

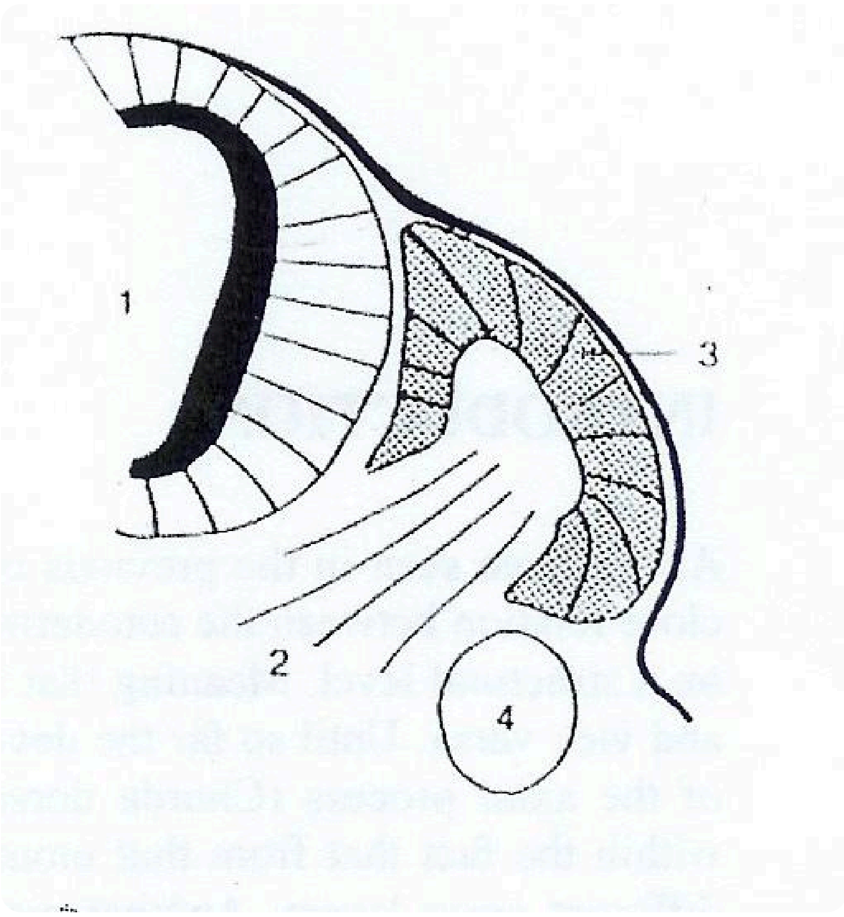

FIG. — Somites and neural tube

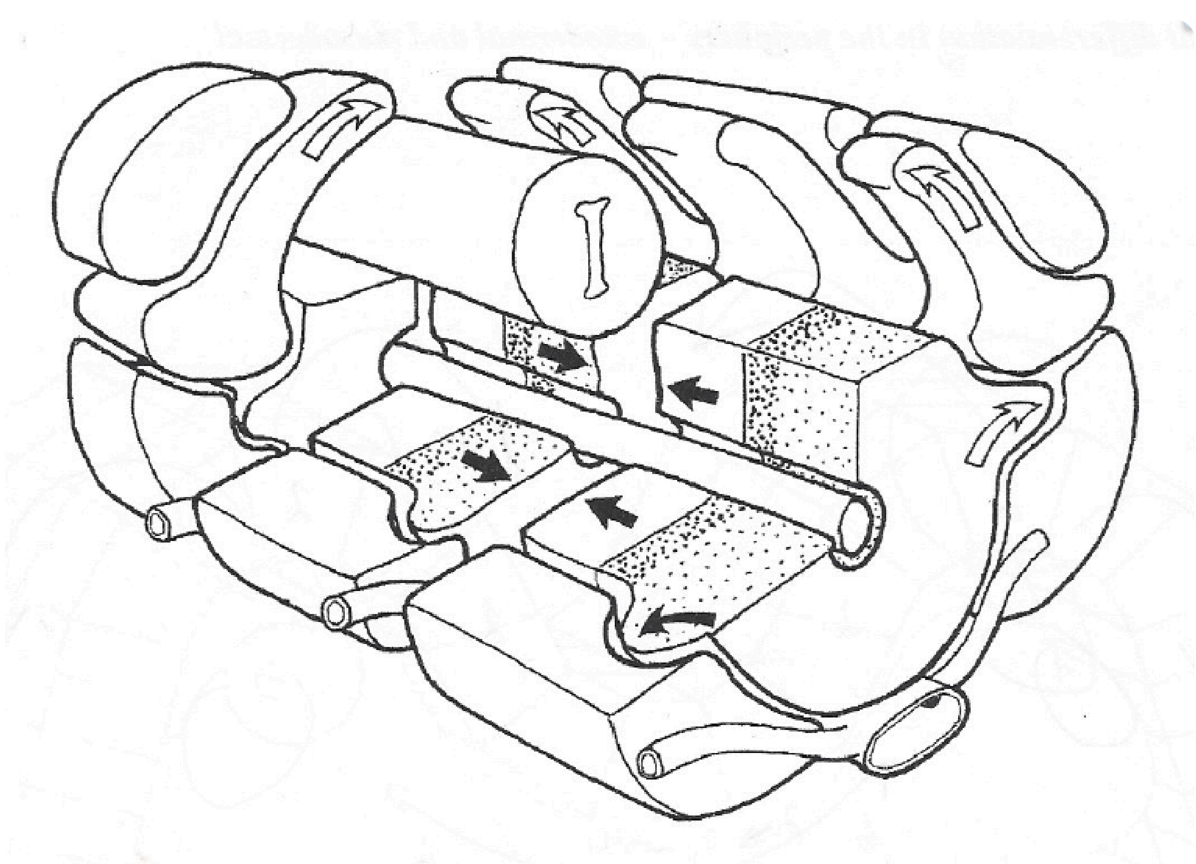

FIG. — Somites and neural tube

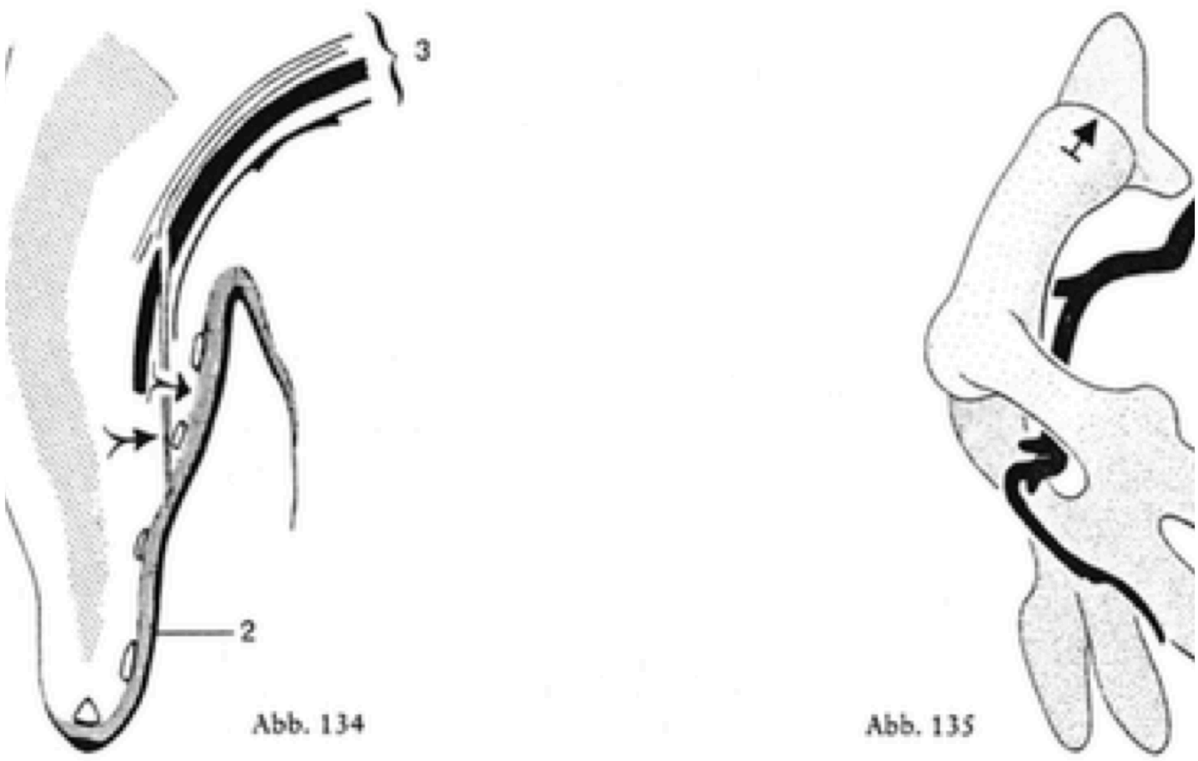

FIG. — Development of the digestive tract